

Multi-dimensional equilibria preserving methods for hyperbolic equations

Davide Torlo, Wasilij Barsukow, Mirco Ciallella, Mario Ricchiuto, Moussa Ziggaf

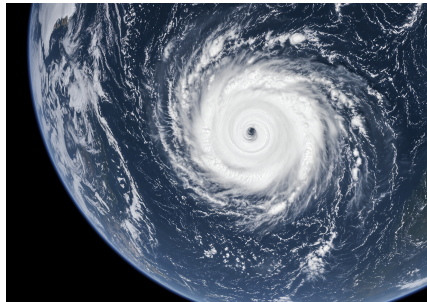
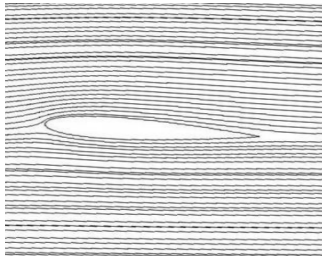
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Moving equilibria in hyperbolic problems



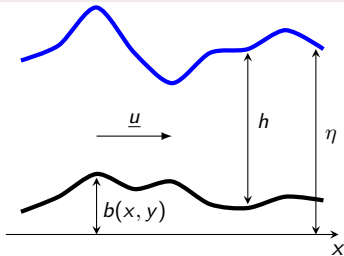
Moving equilibria in hyperbolic problems

Hyperbolic Balance Laws in 2D

$$\partial_t q + \partial_x f(q) + \partial_y g(q) = S(q, x, y, t)$$

Equilibria for Balance Laws in 2D

$$\partial_x f(q) + \partial_y g(q) = S(q, x, y, t)$$



Shallow Water Equations

$$\begin{cases} \partial_t h + \partial_x(hu) + \partial_y(hv) = 0 \\ \partial_t(hu) + \partial_x(hu^2 + \frac{g}{2}h^2) + \partial_y(huv) = -gh\partial_x b \\ \partial_t(hv) + \partial_x(huv) + \partial_y(hv^2 + \frac{g}{2}h^2) = -gh\partial_y b \end{cases}$$

SW equilibria

- Lake at rest: $u = v \equiv 0$, $h + b \equiv \eta_0$ constant
- 1D moving equilibria: $hu \equiv q_0$ constant, h such that $\frac{q_0^2}{2gh^2} + h + b \equiv \eta_0$ constant
- 2D moving equilibria: $\partial_x(hu) + \partial_y(hv) = 0$ and balancing of h , e.g. vortices

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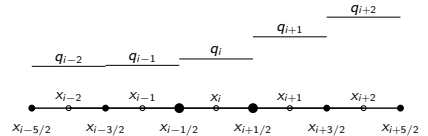
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Finite Volume Method for Hyperbolic Balance Laws

$$\partial_t q + \partial_x f(q) = S(q, x)$$

Finite Volume setting

$$q_i \approx \int_{x_{i-1/2}}^{x_{i+1/2}} q(x) dx$$



Finite Volume Method for Hyperbolic Balance Laws

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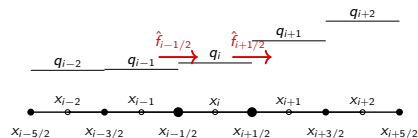
Finite Volume setting

$$q_i \approx \int_{x_{i-1/2}}^{x_{i+1/2}} q(x) dx$$

$$\int_{x_{i-1/2}}^{x_{i+1/2}} q(x, t) + \partial_x f(q) dx = \int_{x_{i-1/2}}^{x_{i+1/2}} S(q, x) dx$$

$$\partial_t q_i + \frac{\hat{f}_{i+1/2} - \hat{f}_{i-1/2}}{\Delta x} = S_i$$

- $\hat{f}_{i+1/2}$ numerical flux (function of $q_{i-k}, \dots, q_{i+\ell+1}$)
- $S_i \approx \int_{x_{i-1/2}}^{x_{i+1/2}} S(q, x) dx$



Finite Volume Method for Hyperbolic Balance Laws

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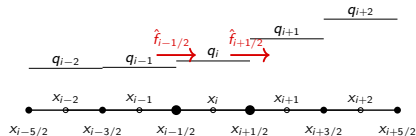
Finite Volume setting

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First order example

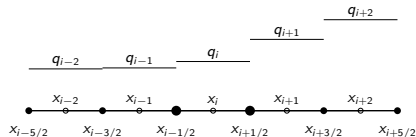
- Upwind flux with positive wind:
 $\hat{f}_{i+1/2} = f(q_i)$
- Rusanov flux:
 $\hat{f}_{i+1/2} = \frac{f(q_i) + f(q_{i+1})}{2} - \frac{\lambda(q_{i+1} - q_i)}{2}$
- Source: $S_i = S(q_i, x_i)$

Global Flux in 1D for FV 1st order¹²³

$$\partial_t q + \partial_x f(q) = S(q, x) \quad \implies \quad \partial_t q + \partial_x G(q, x) = 0$$

$$\text{with } G(q, x) := f(q) - K(q, x) = f(q) - \int_{x_0}^x S(q(s), s) ds.$$

$$\text{FV: } q_i \approx \int_{x_{i-1/2}}^{x_{i+1/2}} q(x) dx$$



¹Gascón, L., Corberán, J. J. Comput. Phys. 172(1), 261–297 (2001)

²Chertock, A., Kurganov, A., Liu, X., Liu, Y., & Wu, T. (2022). Journal of Scientific Computing, 90, 1-21.

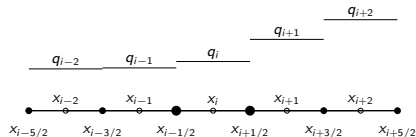
³Ciallella, M., Torlo, D., & Ricchiuto, M. (2023). Journal of Scientific Computing, 96(2), 53.

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$$f_i := \frac{1}{\Delta x} \int_{x_{i-1/2}}^{x_{i+1/2}} f(q(x, t)) dx,$$

$$K_i \approx K(x_i, q(x_i)) = \int_{x_0}^{x_i} S(q(s), s) ds \approx K_{i-1} + \int_{x_{i-1}}^{x_i} S(q(s), s) ds,$$

$$K_i := K_{i-1} + \Delta x \frac{S_{i-1} + S_i}{2}$$

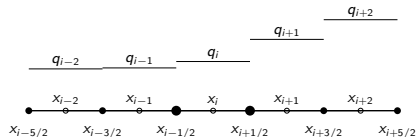
$$G_i := f_i - K_i.$$

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Numerical flux depends only on G :
upwind, Roe, NO Rusanov

$$\partial_t q_i + \frac{\hat{G}_{i+1/2} - \hat{G}_{i-1/2}}{\Delta x} = 0,$$

$$\hat{G}_{i+1/2} := \text{sign}(J)^+ G_i + \text{sign}(J)^- G_{i+1},$$

$$f_i := \frac{1}{\Delta x} \int_{x_{i-1/2}}^{x_{i+1/2}} f(q(x, t)) dx,$$

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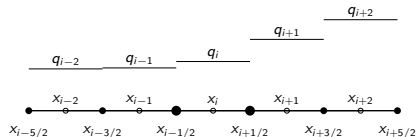
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$$\hat{G}_{i+1/2} := \text{sign}(J)^+ G_i + \text{sign}(J)^- G_{i+1},$$

Equilibrium: $\hat{G}_{i+1/2} = \hat{G}_{i-1/2} = \hat{G}_0$ for all i

$$f_i - K_i = G_0$$

Mind: high order, other equilibria (LAR), super convergence

$$f_i := \frac{1}{\Delta x} \int_{x_{i-1/2}}^{x_{i+1/2}} f(q(x, t)) dx,$$

$$K_i \approx K(x_i, q(x_i)) = \int_{x_0}^{x_i} S(q(s), s) ds \approx K_{i-1} + \int_{x_{i-1/2}}^{x_i} S(q(s), s) ds,$$

$$K_i := K_{i-1} + \Delta x \frac{S_{i-1} + S_i}{2}$$

$$G_i := f_i - K_i.$$

Developing GF 1D FV 1st order

Let's do the computations in a simple case (upwind)!

Formulae

- $\partial_t q_i = -\frac{\hat{G}_{i+1/2} - \hat{G}_{i-1/2}}{\Delta x}$
- $G_i = f_i - K_i$
- $K_i = K_{i-1} + \Delta x \frac{S_{i-1} + S_i}{2}$
- $\text{sign}(J) = +1$
- $\hat{G}_{i+1/2} = G_i$

Classical Upwind FV

$$\partial_t q_i = -\frac{f_i - f_{i-1}}{\Delta x} + S_i$$

Expand!

$$\begin{aligned}\partial_t q_i &= -\frac{\hat{G}_{i+1/2} - \hat{G}_{i-1/2}}{\Delta x} \\ &= -\frac{G_i - G_{i-1}}{\Delta x} \\ &= -\frac{f_i - f_{i-1}}{\Delta x} + \frac{K_i - K_{i-1}}{\Delta x} \\ &= -\frac{f_i - f_{i-1}}{\Delta x} + \frac{S_{i-1} + S_i}{2}.\end{aligned}$$

Validation: Subcritical flow and perturbation

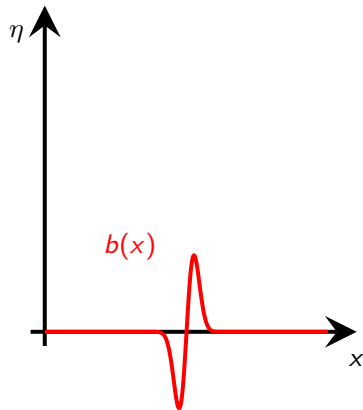
Domain and Bathymetry

$$\begin{aligned}\Omega &= [0, 25], \\ b(x) &= 0.05 \sin(x - 12.5) \exp(1 - (x - 12.5)^2), \\ g &= 9.812.\end{aligned}$$

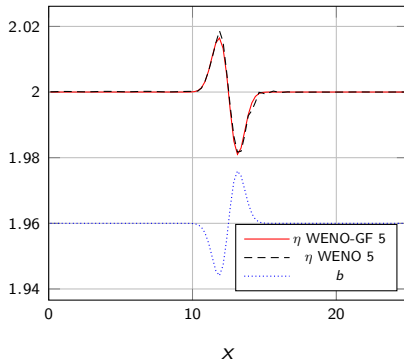
$b(x)$ is chosen C^∞ and such that it has values smaller than machine precision at the boundaries.

Subcritical flow test

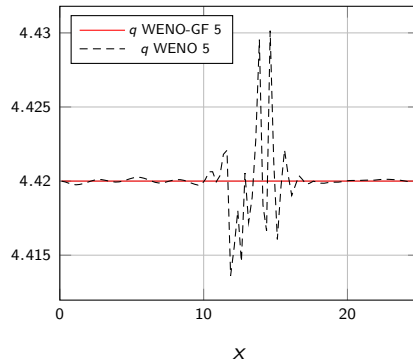
$$\begin{aligned}\text{IC: } h(x, 0) &= 2 - b(x), & q(x, 0) &\equiv 0, \\ \text{BC: } h(25, t) &= 2, & q(0, t) &= 4.42,\end{aligned}$$



Validation: Subcritical flow and perturbation



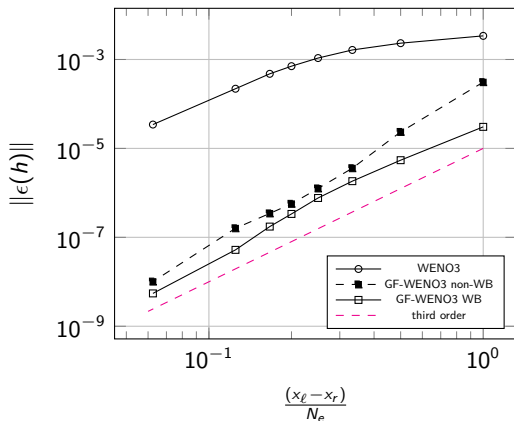
(a) η



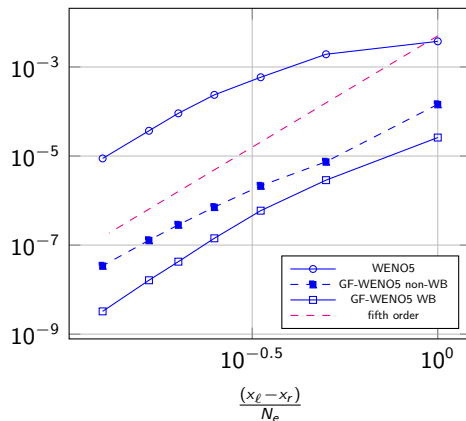
(b) q

Figure: Subcritical flow: characteristic variables compute by means of the GF-WENO5 (red continuous line) and WENO5 (black dashed line) schemes with $N_e = 100$.

Validation: Subcritical flow and perturbation



(a) WENO3



(b) WENO5

Figure: Subcritical flow: convergence tests with WENO3 and WENO5.

Validation: Subcritical flow and perturbation

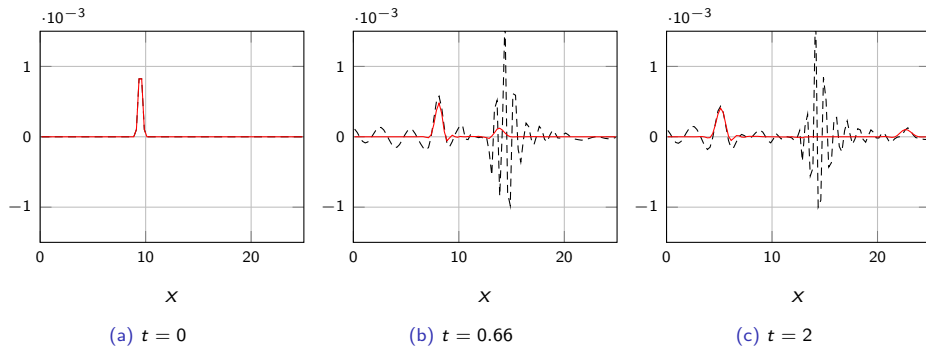


Figure: Perturbation on a subcritical flow: $\eta - \eta^{eq}$

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Acoustics

$$\partial_t u + \partial_x p = S_u$$

$$\partial_t v + \partial_y p = S_v$$

$$\partial_t p + \partial_x u + \partial_y v = S_p$$

SUPG FEM for acoustics

$$\int \varphi(\partial_t u + \partial_x p + S_u) + \alpha \Delta \partial_x \varphi(\partial_t p + \partial_x u + \partial_y v + S_p) = 0$$

$$\int \varphi(\partial_t v + \partial_y p + S_v) + \alpha \Delta \partial_y \varphi(\partial_t p + \partial_x u + \partial_y v + S_p) = 0$$

$$\begin{aligned} \int \varphi(\partial_t p + \partial_x u + \partial_y v + S_p) &+ \alpha \Delta \partial_x \varphi(\partial_t u + \partial_x p + S_u) \\ &+ \alpha \Delta \partial_y \varphi(\partial_t v + \partial_y p + S_v) = 0 \end{aligned}$$

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Acoustics

$$\partial_t \mathbf{u} + \partial_x \mathbf{p} = S_u$$

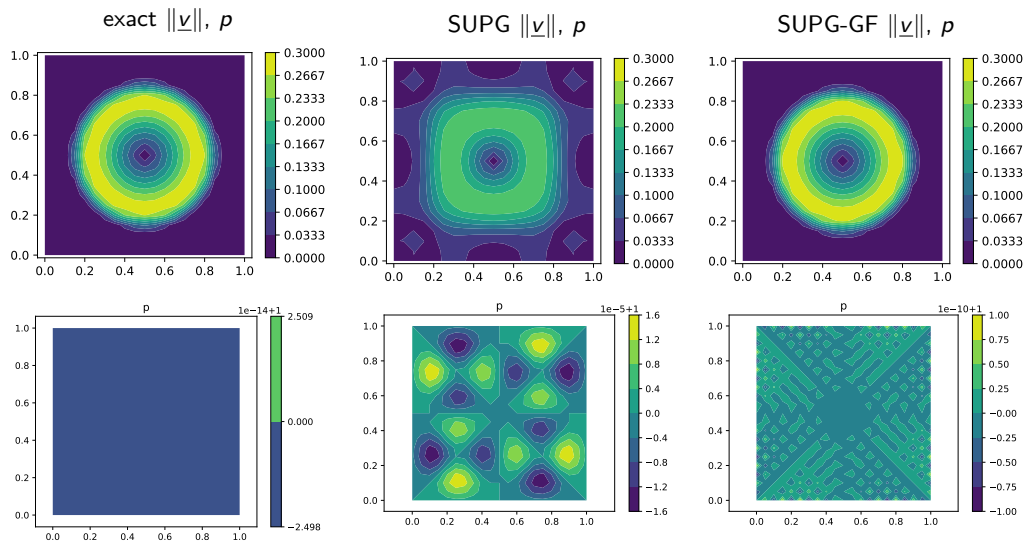
$$\partial_t \mathbf{v} + \partial_y \mathbf{p} = S_v$$

$$\partial_t \mathbf{p} + \partial_x \mathbf{u} + \partial_y \mathbf{v} = S_p$$

Details on discretization

- Cartesian grid!!
- Gauss-Lobatto points for quadrature and Lagrange basis function
- Explicit arbitrary high order time discretization with Deferred Correction

FEM-SUPG $\mathbb{Q}^1$ on an acoustic vortex $S = 0$



Global flux in 2D

Main idea

Define

$$\sigma_x(x, y) := \int_{y_0}^y u(x, s) ds \quad \sigma_y(x, y) := \int_{x_0}^x v(s, y) ds \quad S_p(x, y) := \int_{y_0}^y \int_{x_0}^x S_p(z, s) dz ds,$$

so that

$$\partial_t p + \partial_x u + \partial_y v = \partial_t p + \partial_{xy} \underbrace{(\sigma_x + \sigma_y - S_p)}_{=: \mathcal{F}} = 0$$

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Global Flux SUPG for acoustics

$$\int \varphi(\partial_t u + \partial_x(p - S_u)) + \alpha \Delta x \partial_x \varphi(\partial_t p + \partial_x \partial_y \mathcal{F}) = 0$$

$$\int \varphi(\partial_t v + \partial_y(p - S_v)) + \alpha \Delta y \partial_y \varphi(\partial_t p + \partial_x \partial_y \mathcal{F}) = 0$$

$$\int \varphi(\partial_t p + \partial_x \partial_y \mathcal{F}) + \alpha \Delta x \partial_x \varphi(\partial_t u + \partial_x(p - S_u)) + \alpha \Delta y \partial_y \varphi(\partial_t v + \partial_y(p - S_v)) = 0$$

Global Flux SUPG for acoustics

$$\sigma_x(x, y) := \int_{y_0}^y u(x, s) ds \text{ and } \sigma_y(x, y) := \int_{x_0}^x v(s, y) ds, \text{ with } \sigma_x, \sigma_y \in V_h^K(\Omega_h).$$

Changes in equilibrium

$$\begin{aligned}\nabla \cdot \underline{v} &= 0 \\ \implies \partial_x \partial_y (\sigma_x + \sigma_y) &= 0 \\ \iff \sigma_x + \sigma_y &= f(x) + g(y)\end{aligned}$$

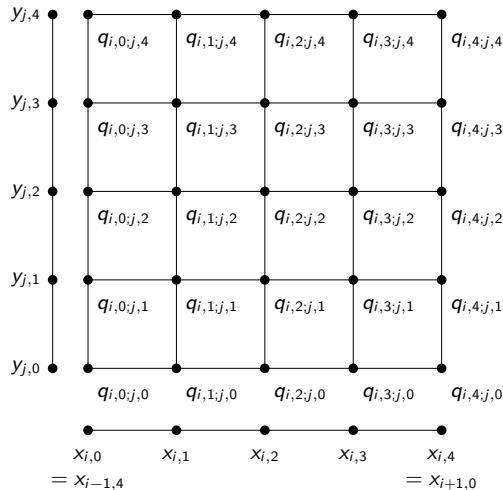
Discrete equilibrium

$$\begin{aligned}\partial_x \partial_y \mathcal{F}(x_i, y_j) &= 0 \\ \implies \int_{x_0}^{x_i} \int_{y_0}^{y_j} \partial_y \partial_x \mathcal{F}(x, y) dx dy &= 0 \quad \forall i, j \\ \implies \int_{x_0}^{x_i} \partial_x \mathcal{F}(x, y_j) dx - \int_{x_0}^{x_i} \partial_x \mathcal{F}(x, y_0) dx &= 0 \quad \forall i, j \\ \implies \mathcal{F}(x_i, y_j) - \mathcal{F}(x_0, y_j) - \mathcal{F}(x_i, y_0) + \mathcal{F}(x_0, y_0) &= 0 \quad \forall i, j \\ \implies \mathcal{F}(x_i, y_j) &= f_i + g_j\end{aligned}$$

Global Flux Definition DoF by DoF in FEM

Definitions of Global Flux

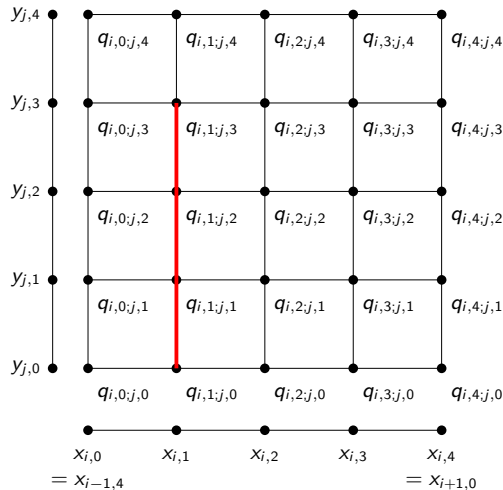
- $V_h^K(\Omega) := \{v \in C(\Omega_h) : v|_E \in \mathbb{Q}^K(E) = \mathbb{P}^K(E_x) \otimes \mathbb{P}^K(E_y), \forall E \in \Omega_h\}$
- $\mathcal{F} := \sigma_x + \sigma_y - \mathcal{S}_p \in V_h^K(\Omega)$
- $\sigma_x(x, y) := \int_{y_0}^y u(x, s) ds$ and
 $\sigma_y(x, y) := \int_{x_0}^x v(s, y) ds$
- $\mathcal{S}_p(x, y) := \int_{y_0}^y \int_{x_0}^x S_p(z, s) dz ds$
- $\sigma_x, \sigma_y, \mathcal{S}_p \in V_h^K$



Global Flux Definition DoF by DoF in FEM

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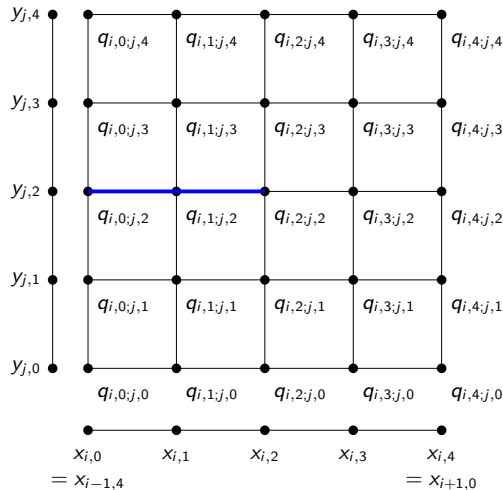
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- $\sigma_x, \sigma_y, \mathcal{S}_p \in V_h^K$
- $\sigma_x(x_{i,k}, y_{j,\ell}) = \int_{y_{j,0}}^{y_{j,\ell}} u(x_{i,k}, s) ds$



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Definitions of Global Flux

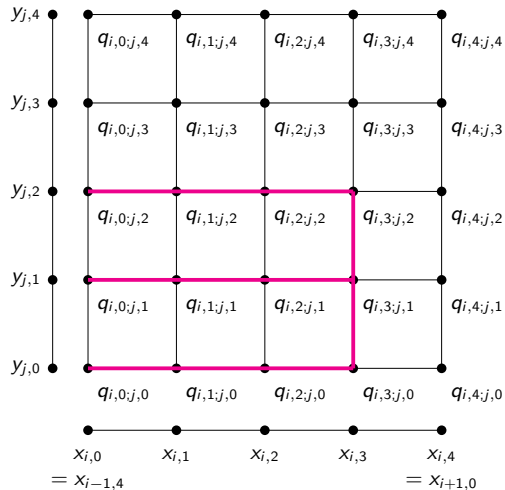
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- $\sigma_y(x_{i,k}, y_{j,\ell}) = \int_{x_{i,0}}^{x_{i,k}} v(s, y_{j,\ell}) ds$



Global Flux Definition DoF by DoF in FEM

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- $V_h^K(\Omega) := \{v \in C(\Omega_h) : v|_E \in \mathbb{Q}^K(E) = \mathbb{P}^K(E_x) \otimes \mathbb{P}^K(E_y), \forall E \in \Omega_h\}$
- $\mathcal{F} := \sigma_x + \sigma_y - \mathcal{S}_p \in V_h^K(\Omega)$
- $\sigma_x(x, y) := \int_{y_0}^y u(x, s) ds$ and
 $\sigma_y(x, y) := \int_{x_0}^x v(s, y) ds$
- $\mathcal{S}_p(x, y) := \int_{y_0}^y \int_{x_0}^x S_p(z, s) dz ds$
- $\sigma_x, \sigma_y, \mathcal{S}_p \in V_h^K$
- $\sigma_x(x_{i,k}, y_{j,\ell}) = \int_{y_{j,0}}^{y_{j,\ell}} u(x_{i,k}, s) ds$
- $\sigma_y(x_{i,k}, y_{j,\ell}) = \int_{x_{i,0}}^{x_{i,k}} v(s, y_{j,\ell}) ds$
- $\mathcal{S}_p(x_{i,k}, y_{j,\ell}) = \int_{y_{j,0}}^{y_{j,\ell}} \int_{x_{i,0}}^{x_{i,k}} S_p(z, s) dz ds$



Global Flux is not global!

- In principle $\sigma_x(x, y) = \int_{y_B}^y u(x, s) ds$ should be integrated from the beginning (bottom) of the domain y_B !
- In practice we always use $\partial_x \partial_y \sigma_x(x, y)$ integrated in one cell!!!!
- So,

$$\sigma_x(x, y) = \int_{y_B}^y u(x, s) ds = \underbrace{\int_{y_B}^{y_0} u(x, s) ds}_{\text{constant in one cell!}} + \int_{y_0}^y u(x, s) ds$$

whatever constant we bring from outside the cell, is canceled out

$$\partial_y \sigma_x(x, y) = \partial_y \int_{y_B}^y u(x, s) ds = \partial_y \int_{y_B}^{y_0} u(x, s) ds + \partial_y \int_{y_0}^y u(x, s) ds = \partial_y \int_{y_0}^y u(x, s) ds$$

- At the discrete level we have an integral operator I_y and a differential operator D_y that together give a weird averaging operator $D_y I_y$

Analogy with Raviart-Thomas elements

Global Flux velocity space

Recall that $u, v \in \mathbb{P}^d \otimes \mathbb{P}^d$ continuous across interfaces. Then, we define

$$\begin{aligned} \int^y u(x, s) ds &\in \mathbb{P}^d \otimes \mathbb{P}^{d+1} & \sigma_x &= \Pi_{\mathbb{P}^d \otimes \mathbb{P}^d} \left(\int^y u(x, s) ds \right) \in \mathbb{P}^d \otimes \mathbb{P}^d, \\ \tilde{u} &:= \partial_y \sigma_x \in \mathbb{P}^d \otimes d\mathbb{P}^{d-1}, & \tilde{v} &:= \partial_x \sigma_y \in \mathbb{P}^{d-1} \otimes d\mathbb{P}^d. \end{aligned}$$

Now, the equation that we actually discretize is $\partial_t p + \partial_x \tilde{u} + \partial_y \tilde{v} = 0$ for the pressure p . Here,

$$(\tilde{u}, \tilde{v}) \in \mathbb{P}^d \otimes d\mathbb{P}^{d-1} \times d\mathbb{P}^{d-1} \otimes \mathbb{P}^d = \text{RT}^d.$$

Differences with RT

- We evolve anyway u, v in $\mathbb{P}^d \otimes \mathbb{P}^d$ and we do not need to evolve $\tilde{u}, \tilde{v}$ in RT space
- We use a non-standard projection (defined DoF by DoF with integration)
- The SUPG stabilization is still necessary (we run explicit methods)
- We can include sources

Shift of paradigm for kernels

Since we have written the SUPG-GF as

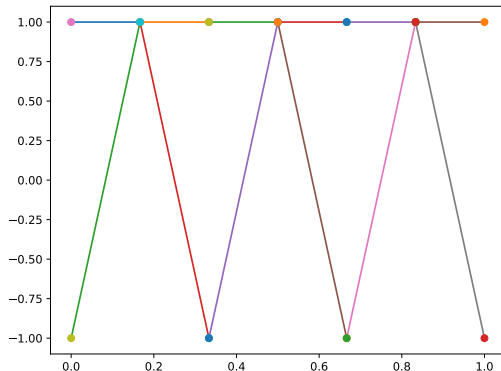
$$\int_{\Omega} (\varphi + \alpha \Delta \partial_x \varphi J^x + \alpha \Delta \partial_y \varphi J^y) (\partial_t \mathbf{q} + \partial_{xy} \mathcal{F}) = 0 \quad \forall \varphi,$$

we need to study the operators given by

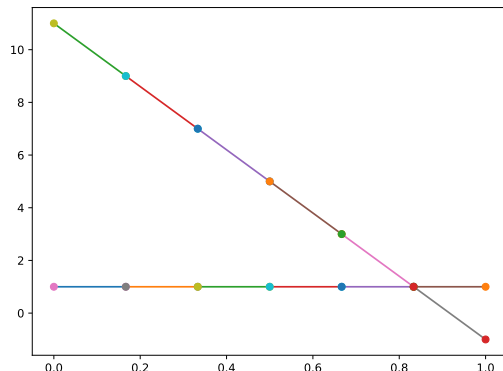
$$D_x \otimes D_y := \int_{\Omega} \varphi_i \partial_{xy} \varphi_j, \quad D_x^* \otimes D_y := \int_{\Omega} \partial_x \varphi_i \partial_{xy} \varphi_j, \quad D_x \otimes D_y^* := \int_{\Omega} \partial_y \varphi_i \partial_{xy} \varphi_j.$$

One dimensional kernels of D_x and $D_x^\times$

Kernel of D_x , 2 basis functions



Kernel of $D_x^\times$, 2 basis functions



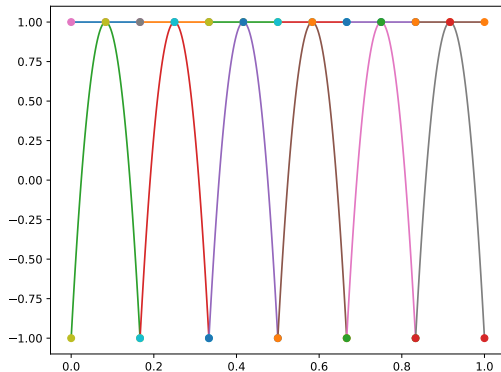
Operators

• Divergence $D_x \otimes D_y$

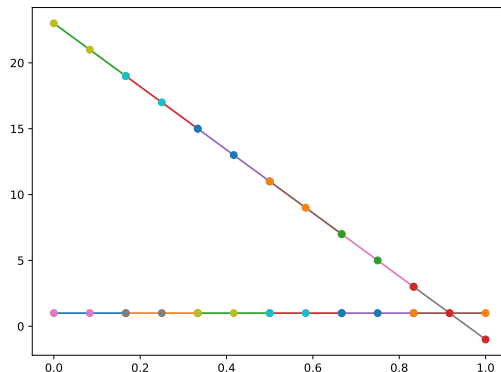
• Stabilization $D_x^\times \otimes D_y, D_x \otimes D_y^\times$

One dimensional kernels of D_x and $D_x^\times$

Kernel of D_x , 2 basis functions



Kernel of $D_x^\times$, 2 basis functions



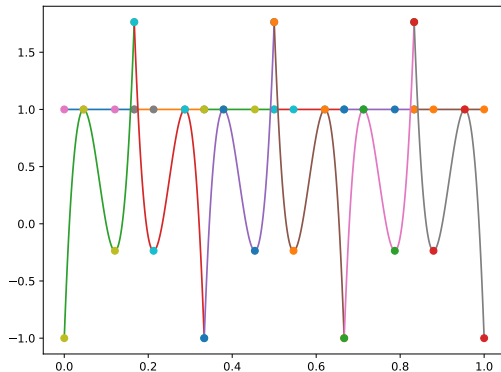
Operators

• Divergence $D_x \otimes D_y$

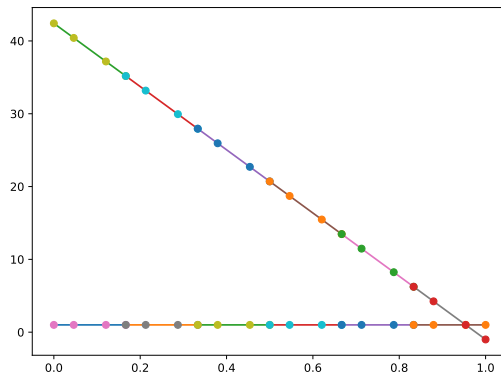
• Stabilization $D_x^\times \otimes D_y, D_x \otimes D_y^\times$

One dimensional kernels of D_x and $D_x^\times$

Kernel of D_x , 2 basis functions



Kernel of $D_x^\times$, 2 basis functions



Operators

• Divergence $D_x \otimes D_y$

• Stabilization $D_x^\times \otimes D_y, D_x \otimes D_y^\times$

Kernels of D_x , $D_x^\times$ and 2D operators

Theorem

- $\ker(D_x) = \text{span}\{1, w\}$ with w with first derivative discontinuous at each interface (chainsaw function)
- $\ker(D_x^\times) = \text{span}\{1, x\}$

Kernels of D_x , $D_x^\times$ and 2D operators

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Corollary

- $w \notin \ker(D_x^\times) \Rightarrow$ if stable $D_x^\times$ should dissipate spurious oscillating modes that are in the kernel of D_x

Kernels of D_x , D_x^x and 2D operators

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Kernel of $D_x \otimes D_y$

- $\ker(D_x \otimes D_y) = \ker(D_x) \otimes \mathbb{R}^{N_y K+1} + \mathbb{R}^{N_x K+1} \otimes \ker(D_y)$

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- $D_x \otimes D_y \mathcal{F} = 0 \iff \mathcal{F} = \Phi_x + \Phi_y$ with

Kernels of D_x , D_x^x and 2D operators

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- $\Phi_y \in \mathbb{R}^{N_x K+1} \otimes \ker(D_y)$
- $\Phi_x = \mathbf{1} \otimes g_1 + w \otimes g_2$ with $g_1, g_2 \in \mathbb{R}^{N_y K+1}$
- $\Phi_y = f_1 \otimes \mathbf{1} + f_2 \otimes w$ with $f_1, f_2 \in \mathbb{R}^{N_x K+1}$

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- $\Phi_y = f_1 \otimes 1 + f_2 \otimes w$ with $f_1, f_2 \in \mathbb{R}^{N_x K+1}$

Corollary

- $w \notin \ker(D_x^x) \Rightarrow$ if stable D_x^x should dissipate spurious oscillating modes that are in the kernel of D_x

Two Dimensional Operators

- $\Phi_x = \underbrace{1 \otimes g_1}_{\text{Good equilibrium}} + \underbrace{w \otimes g_2}_{\text{Bad equilibrium}}$

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- $(D_x^\times \otimes D_y)(1 \otimes g_1) = D_x^\times 1 \otimes D_y g_1 = 0 \otimes * = 0$
- $(D_x \otimes D_y^\times)(1 \otimes g_1) = D_x 1 \otimes D_y^\times g_1 = 0 \otimes * = 0$

Kernels of D_x , D_x^x and 2D operators

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- $(D_x \otimes D_y^y)(1 \otimes g_1) = D_x 1 \otimes D_y^y g_1 = 0 \otimes * = 0$
- $(D_x \otimes D_y^y)(w \otimes g_2) = D_x w \otimes D_y^y g_2 = 0 \otimes * = 0$
- $(D_x^x \otimes D_y)(w \otimes g_2) = D_x^x w \otimes D_y g_2 \neq 0$ except
- $(D_x^x \otimes D_y)(w \otimes 1) = D_x^x w \otimes D_y 1 = 0$ and
- $(D_x^x \otimes D_y)(w \otimes w) = D_x^x w \otimes D_y w = 0$

GF+FEM+SUPG

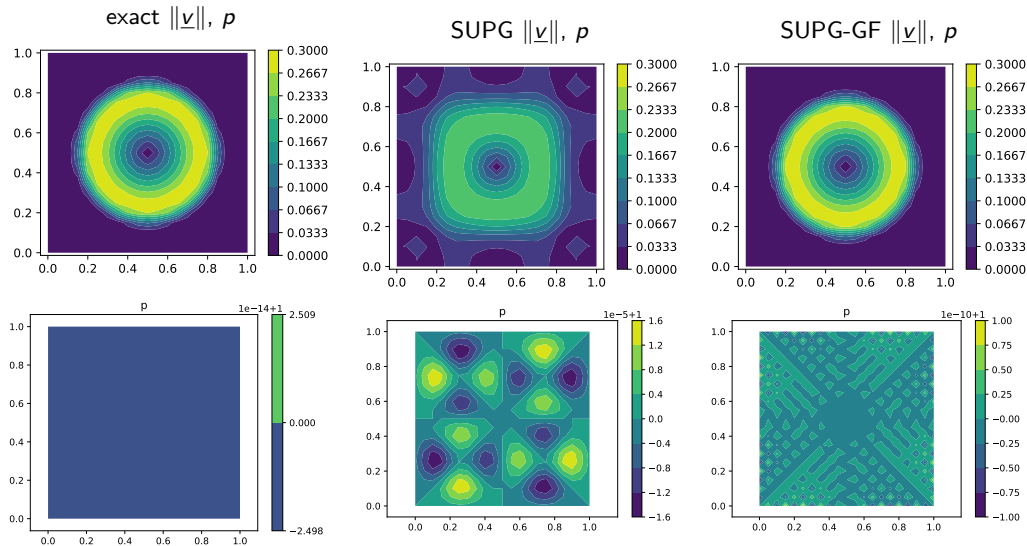
$$\partial_t \mathbf{q} + \partial_x f(\mathbf{q}) + \partial_y g(\mathbf{q}) = S(\mathbf{q}) \implies \partial_t \mathbf{q} + \partial_x \partial_y \mathcal{F} = 0$$

$$\mathcal{F} := \int^y f(\mathbf{q}) + \int^x g(\mathbf{q}) - \int^x \int^y S(\mathbf{q});$$

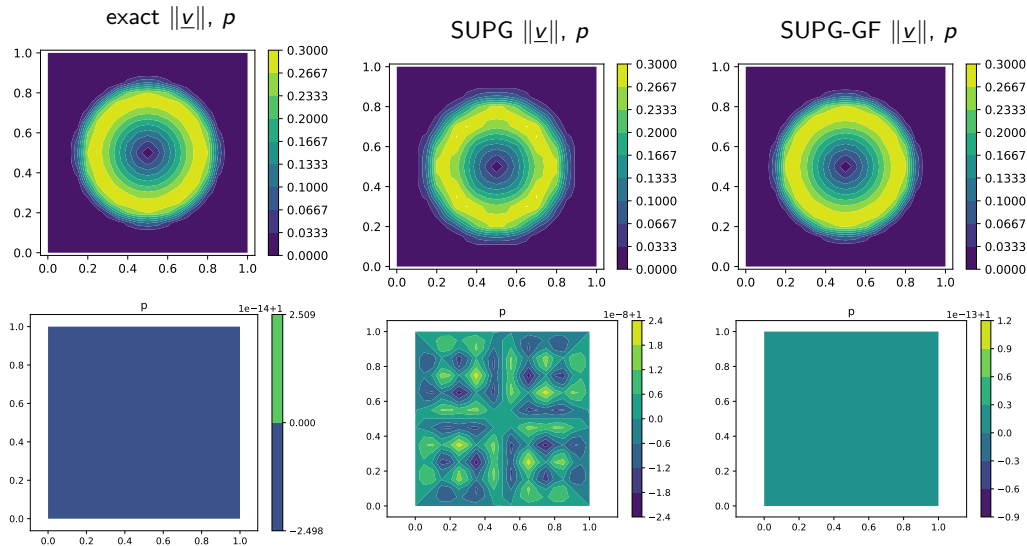
$$\int_{\Omega} (\varphi + \alpha \Delta \partial_x \varphi J^x + \alpha \Delta \partial_y \varphi J^y) (\partial_t \mathbf{q} + \partial_{xy} \mathcal{F}) = 0 \quad \forall \varphi.$$

- J^x and J^y are the Jacobians of the fluxes f and g .

Simulation of vortex (linear acoustics) FEM+SUPG: $\mathbb{Q}^1$, $N_x = N_y = 20$



Simulation of vortex (linear acoustics) FEM+SUPG: $\mathbb{Q}^2$, $N_x = N_y = 10$



Simulation of vortex: errors

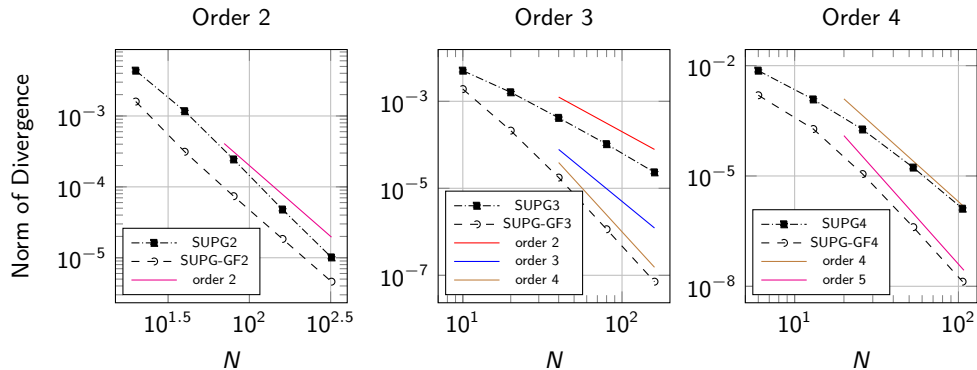


Figure: Smooth vortex: convergence of L^2 error of u with respect to the number of elements in x

Pressure perturbation

- Gaussian centered in $\underline{x}_p = (0.4, 0.43)$
- scaling coefficient $r_0 = 0.1$
- radius $\rho(\underline{x}) = \sqrt{\|\underline{x} - \underline{x}_p\|}/r_0$

$$\delta_p(\underline{x}) = \varepsilon e^{-\frac{1}{2(1-\rho(\underline{x}))^2} + \frac{1}{2}},$$

- final time $T = 0.35$

Vortex perturbation

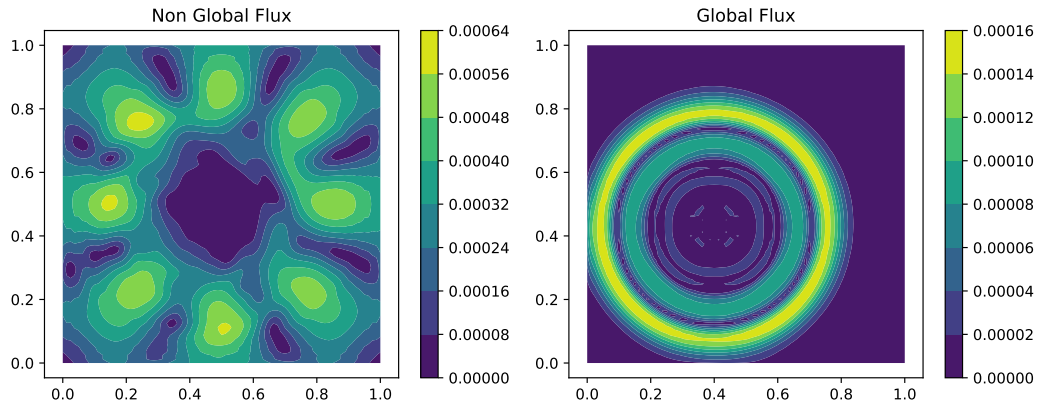


Figure: Perturbation($\varepsilon = 10^{-3}$) test. Plot of $\|\underline{u}_{eq} - \underline{u}_p\|$, with $\underline{u}_{eq}$ the equilibrium obtained with a cheap optimization process. $\mathbb{P}^1$ with 80×80 cells and 6561 dofs.

Vortex perturbation

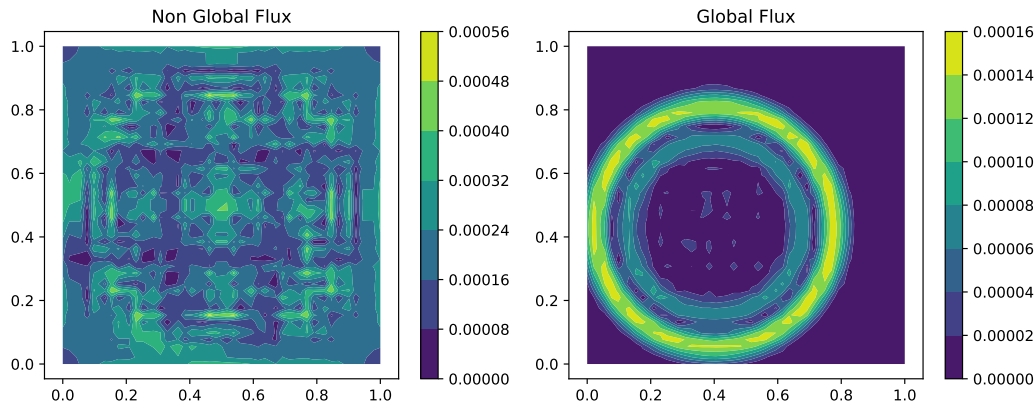


Figure: Perturbation($\varepsilon = 10^{-3}$) test. Plot of $\|\underline{u}_{eq} - \underline{u}_p\|$, with $\underline{u}_{eq}$ the equilibrium obtained with a cheap optimization process. $\mathbb{P}^3$ with 13×13 cells and 1600 dofs.

Vortex perturbation

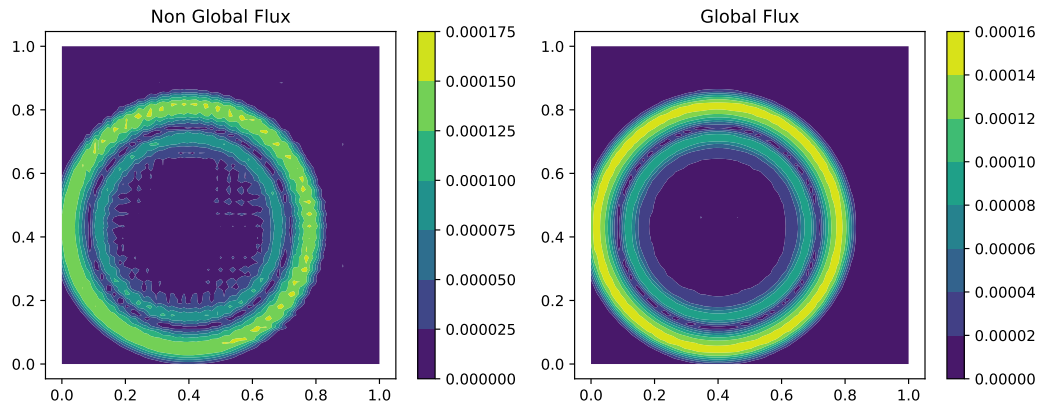
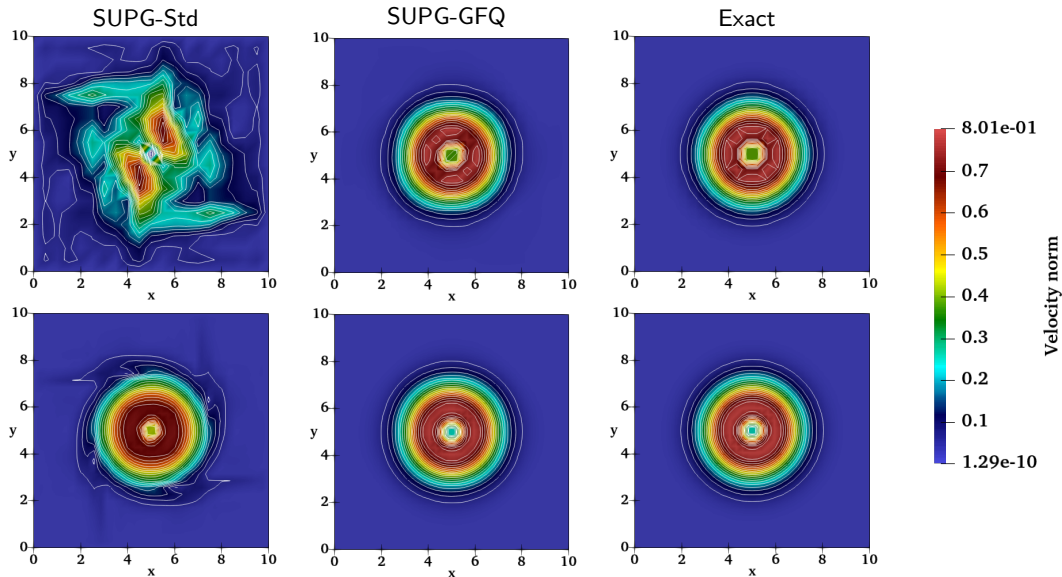


Figure: Perturbation($\varepsilon = 10^{-3}$) test. Plot of $\|\underline{u}_{eq} - \underline{u}_p\|$, with $\underline{u}_{eq}$ the equilibrium obtained with a cheap optimization process. $\mathbb{P}^3$ with 26 cells and 6241 dofs.

Euler equations: isentropic steady vortex Mach = 0.7 $t = 50 \mathbb{Q}^1$ $N = 15, 35$ (top,bot)



Euler equations: isentropic steady vortex (superconvergence and low Mach)

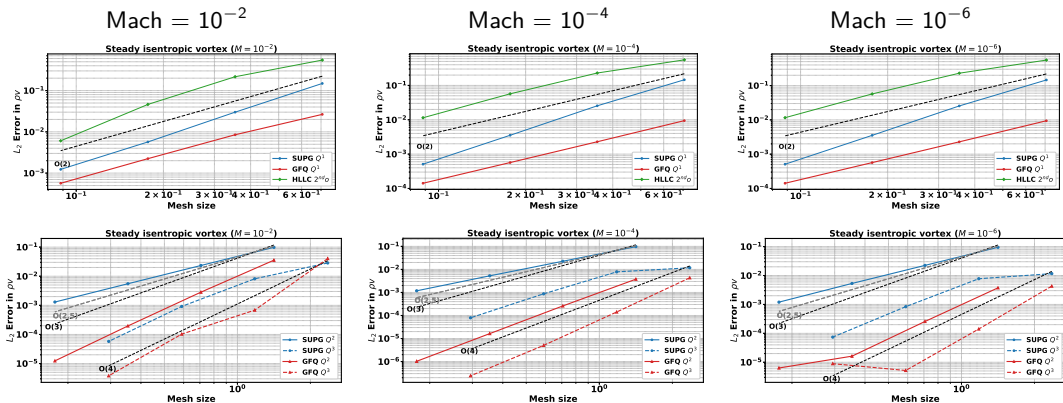
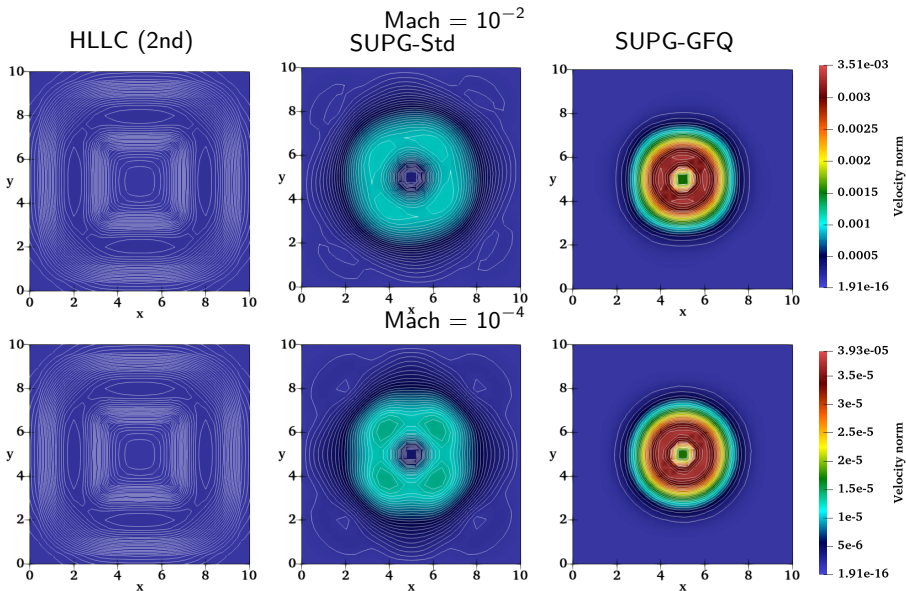
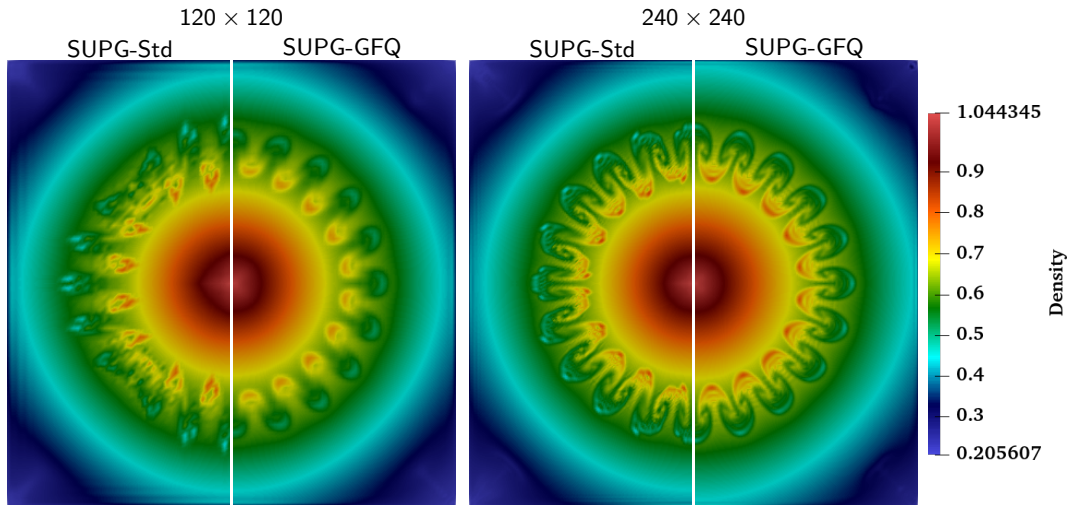


Figure: Convergence of the relative error on ρv for Q^1 SUPG-Std and SUPG-GFQ and for FV-HLLC (left) and Q^2 and Q^3 (right) at time 50 for the steady isentropic vortex test case with Mach numbers $M = 10^{-2}$ (**top**), $M = 10^{-4}$ (**center**) and $M = 10^{-6}$ (**bottom**).

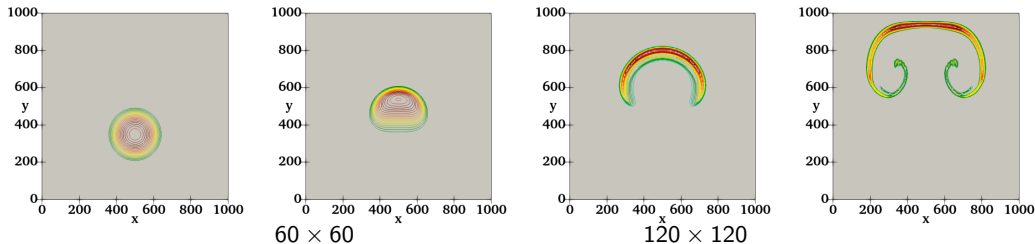
Euler equations: isentropic steady vortex (low Mach $\mathbb{Q}^1$ $N = 25$ $t = 2000$)



Euler with radial gravity: Rayleigh-Taylor instability



Euler equations: Thermal rising bubble Q^1 120×120

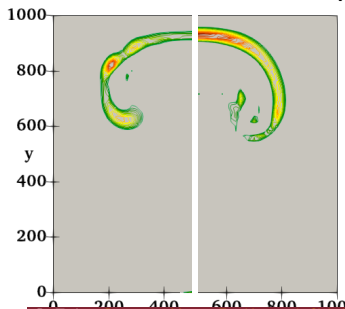


Top: GF
150x150

Bottom
left: GF
60x60

Bottom
right:
SUPG
60x60

SUPG-Std SUPG-GFQ



SUPG-Std SUPG-GFQ

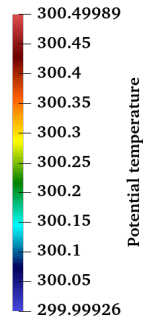
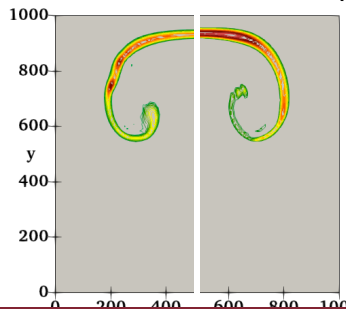


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Finite Volume Corner-Flux Formulation

Equation

$$\partial_t q(t, x, y) + \partial_x \partial_y \mathcal{F}(t, x, y) = 0,$$

Finite Volume Formulation

$$\frac{q_{i,j}^{n+1} - q_{i,j}^n}{\Delta t} + \frac{\mathcal{F}(t^n, x_{i+\frac{1}{2}}, y_{j+\frac{1}{2}}) - \mathcal{F}(t^n, x_{i-\frac{1}{2}}, y_{j+\frac{1}{2}}) - \mathcal{F}(t^n, x_{i+\frac{1}{2}}, y_{j-\frac{1}{2}}) + \mathcal{F}(t^n, x_{i-\frac{1}{2}}, y_{j-\frac{1}{2}})}{\Delta x \Delta y} = 0. \quad (1)$$

$$\widehat{\mathcal{F}}_{i+\frac{1}{2}, j+\frac{1}{2}}^{(i,j)} \approx \pm \mathcal{F}(t^n, x_{i+\frac{1}{2}}, y_{j+\frac{1}{2}}) \quad (2)$$

Corner flux definition¹

$$\frac{q_{i,j}^{n+1} - q_{i,j}^n}{\Delta t} + \frac{\widehat{\mathcal{F}}_{i+\frac{1}{2}, j+\frac{1}{2}}^{(i,j)} + \widehat{\mathcal{F}}_{i+\frac{1}{2}, j+\frac{1}{2}}^{(i+1,j+1)} + \widehat{\mathcal{F}}_{i+\frac{1}{2}, j+\frac{1}{2}}^{(i+1,j)} + \widehat{\mathcal{F}}_{i+\frac{1}{2}, j+\frac{1}{2}}^{(i,j+1)}}{\Delta x \Delta y} = 0. \quad (3)$$

¹Barsukow, Ciallella, Riuichiuto, Torlo. 2026.

Corner Flux Sketch (Dual Cell SUPG formulation)

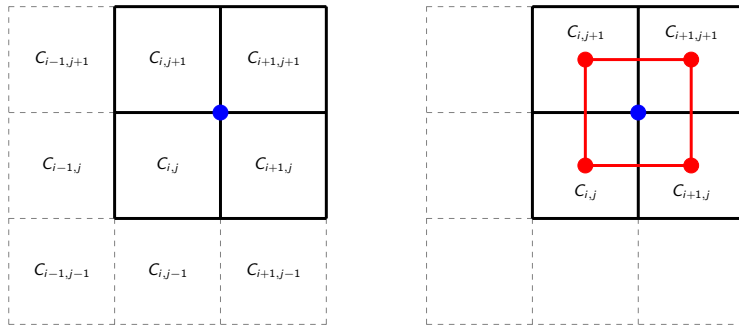


Figure: Cell labeling for the 2D grid: main grid (left), dual grid (right).

- SUPG Corner Fluxes

$$\begin{aligned}\widehat{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j+1)} &= \overline{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}} n_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j+1)} + \mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j+1)}, & \widehat{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j+1)} &= \overline{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}} n_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j+1)} + \mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j+1)}, \\ \widehat{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j)} &= \overline{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}} n_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j)} + \mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j)}, & \widehat{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j)} &= \overline{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}} n_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j)} + \mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j)},\end{aligned}$$

- Central Part

$$\overline{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}} = \frac{1}{4}(\mathcal{F}_{i,j} + \mathcal{F}_{i+1,j} + \mathcal{F}_{i+1,j+1} + \mathcal{F}_{i,j+1})$$

- Diffusion Part

$$\begin{aligned}\mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j+1)} &= \frac{\alpha\Delta}{4} \left(-\frac{J^x}{\Delta x} + \frac{J^y}{\Delta y} \right) \Phi_{i+\frac{1}{2},j+\frac{1}{2}}, & \mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j+1)} &= \frac{\alpha\Delta}{4} \left(+\frac{J^x}{\Delta x} + \frac{J^y}{\Delta y} \right) \Phi_{i+\frac{1}{2},j+\frac{1}{2}}, \\ \mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j)} &= \frac{\alpha\Delta}{4} \left(-\frac{J^x}{\Delta x} - \frac{J^y}{\Delta y} \right) \Phi_{i+\frac{1}{2},j+\frac{1}{2}}, & \mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j)} &= \frac{\alpha\Delta}{4} \left(+\frac{J^x}{\Delta x} - \frac{J^y}{\Delta y} \right) \Phi_{i+\frac{1}{2},j+\frac{1}{2}},\end{aligned}$$

$$\Phi_{i+\frac{1}{2},j+\frac{1}{2}} := \Phi(\widetilde{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}}) := \mathcal{F}_{i+1,j+1} - \mathcal{F}_{i,j+1} - \mathcal{F}_{i+1,j} + \mathcal{F}_{i,j} = \int_{\widetilde{\mathcal{C}}_{i+\frac{1}{2},j+\frac{1}{2}}} \partial_{xy} \widetilde{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}} dx dy.$$

- When $\mathcal{F}_{i,j} = \sigma_i + \gamma_j$ **everything cancels out!**

Comparison with Rusanov Corner Flux

- SUPG Corner Fluxes

$$\begin{aligned}\widehat{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j+1)} &= \overline{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}} n_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j+1)} + \mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j+1)}, & \widehat{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j+1)} &= \overline{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}} n_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j+1)} + \mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j+1)}, \\ \widehat{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j)} &= \overline{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}} n_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j)} + \mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j)}, & \widehat{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j)} &= \overline{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}} n_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j)} + \mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j)},\end{aligned}$$

- Central Part

$$\overline{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}} = \frac{1}{4}(\mathcal{F}_{i,j} + \mathcal{F}_{i+1,j} + \mathcal{F}_{i+1,j+1} + \mathcal{F}_{i,j+1})$$

- Diffusion Part

$$\begin{aligned}\mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j+1)} &= \alpha \Delta(q_{i,j+1} - \bar{q}_{i+\frac{1}{2},j+\frac{1}{2}}), & \mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j+1)} &= \alpha \Delta(q_{i+1,j+1} - \bar{q}_{i+\frac{1}{2},j+\frac{1}{2}}), \\ \mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j)} &= \alpha \Delta(q_{i,j} - \bar{q}_{i+\frac{1}{2},j+\frac{1}{2}}), & \mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j)} &= \alpha \Delta(q_{i+1,j} - \bar{q}_{i+\frac{1}{2},j+\frac{1}{2}}), \\ \bar{q}_{i+\frac{1}{2},j+\frac{1}{2}} &= \frac{1}{4}(q_{i,j} + q_{i+1,j} + q_{i+1,j+1} + q_{i,j+1})\end{aligned}$$

- When $\mathcal{F}_{i,j} = \sigma_i + \gamma_j$ **the diffusion part might still contribute!**

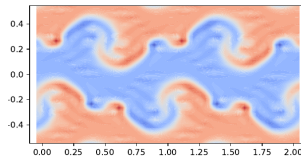
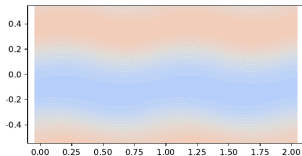
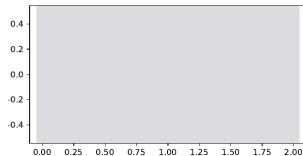
Euler equations: Kevin-Helmoltz instability

$N_x = 64, N_y = 32$

FV-1

FV-2

GF

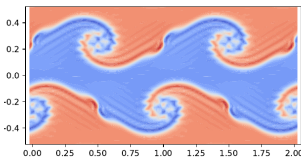
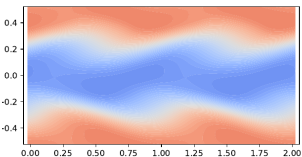
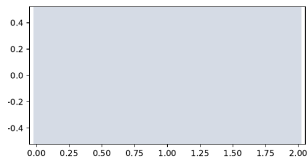


$N_x = 128, N_y = 64$

FV-1

FV-2

GF



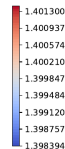
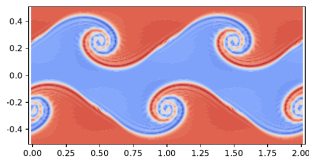
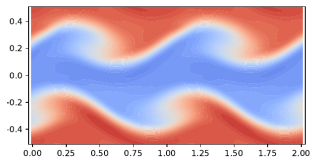
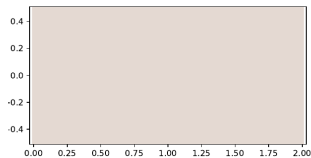
Euler equations: Kevin-Helmoltz instability

$N_x = 256, N_y = 128$

FV-1

FV-2

GF

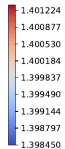
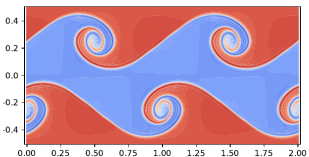
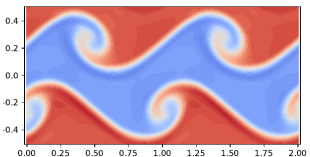
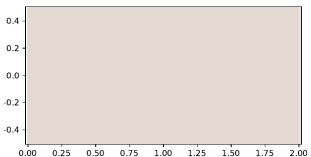


$N_x = 512, N_y = 256$

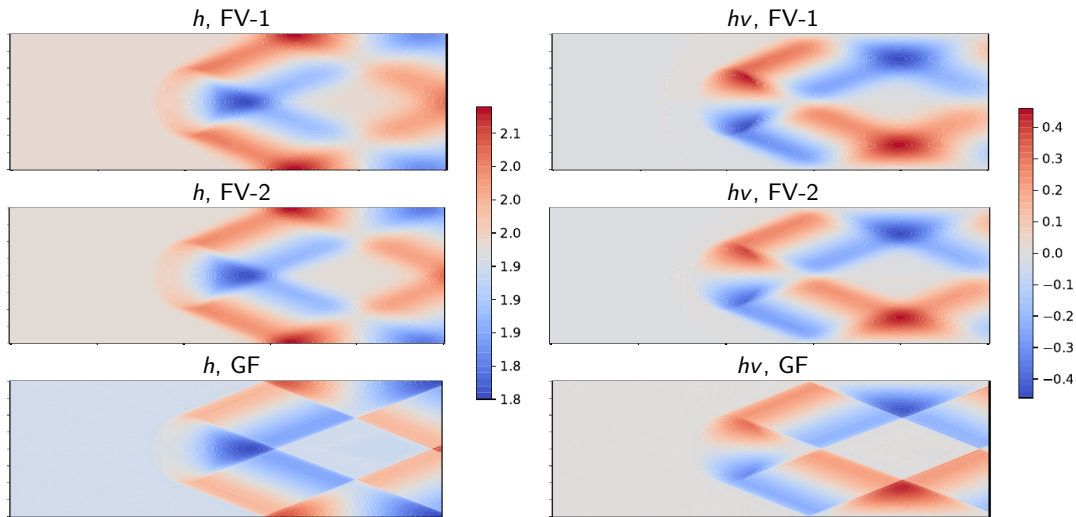
FV-1

FV-2

GF



Shallow water: subcritical flow with bathymetry



Shallow water: subcritical flow with bathymetry

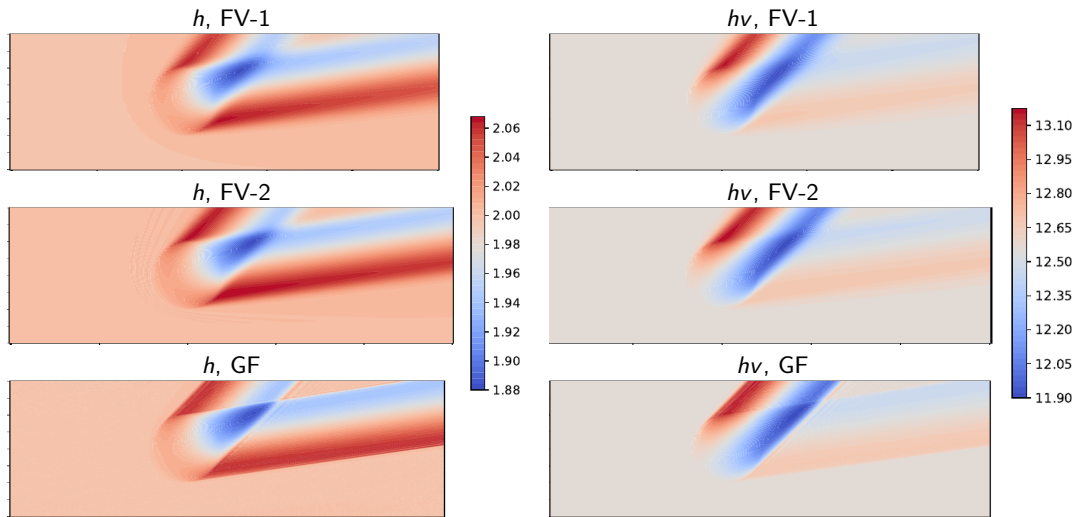
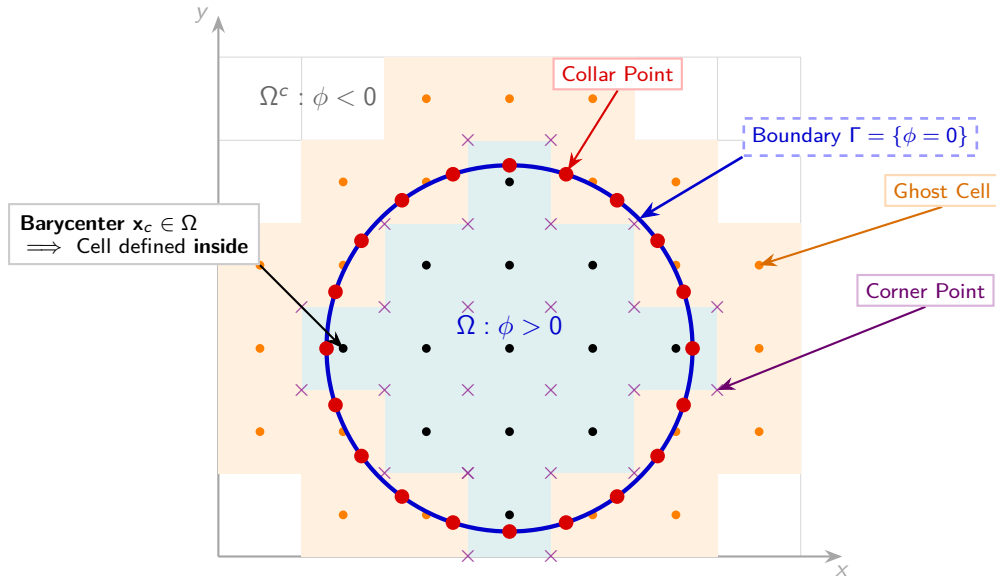


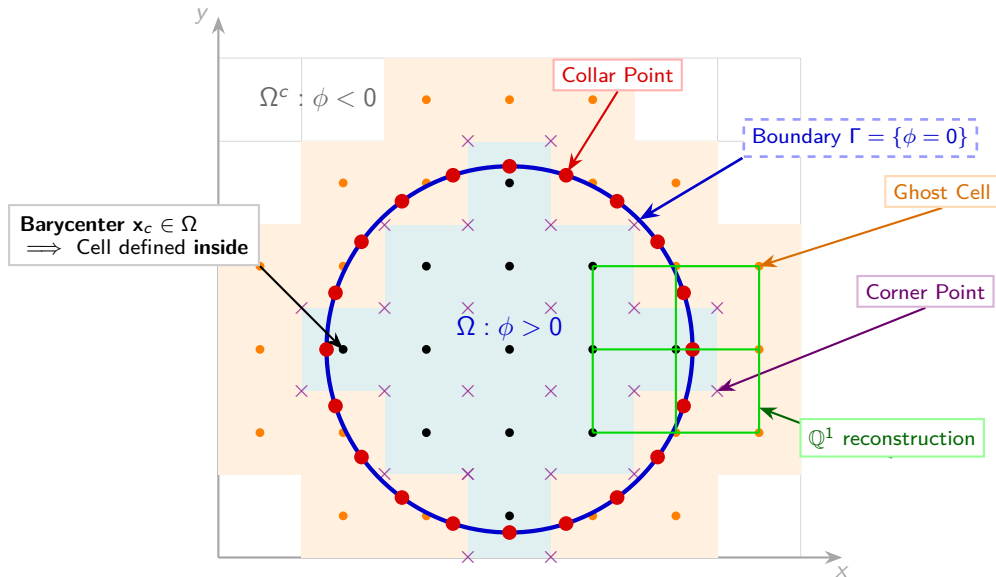
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Geometric Classifications: How to set ghosts cells?



Geometric Classifications: How to set ghosts cells?



Extrapolation from the Interior and Collar Points

Least Square Problem (Dirichlet BC)

$$\min_{p(x,y) \in \mathbb{Q}^1} \sum_{\ell=-1}^1 \sum_{r=-1}^1 (p(x_{i+\ell}^r, y_{j+r}^r) - q(x_{i+\ell}^r, y_{j+r}^r)) \chi(x_{i+\ell}^r, y_{j+r}^r)_{\tilde{\Omega}} \quad \text{subject to} \quad (4)$$
$$p(x_k^c, y_k^c) = u(x_k^c, y_k^c), \quad (x_k^c, y_k^c) \in \Gamma \text{ collar points for } k = 1, \dots, N_{i+\frac{1}{2}, j+\frac{1}{2}}^c,$$

where (x_k^c, y_k^c) are the closest $N_{i+\frac{1}{2}, j+\frac{1}{2}}^c \in \mathbb{N}$ collar points to the corner point $(x_{i+\frac{1}{2}}, y_{j+\frac{1}{2}})$ and $\chi(x_{i+\ell}^r, y_{j+r}^r)_{\tilde{\Omega}}$ is the indicator function that is 1 if the cell with barycenter $(x_{i+\ell}^r, y_{j+r}^r)$ is in the computational domain and 0 otherwise.

Computational Aspects

- Lagrange Multiplier Method
- Least Square Matrices are small
- Can be computed at the beginning of the simulation (and inverted)
- Number of collar points chosen to stabilize and make solvable the system
- Cheap
- **Second Order Accurate**

Neumann BC

$$\min_{p(x,y) \in \mathbb{Q}^1} \sum_{\ell=-1}^1 \sum_{r=-1}^1 (p(x_{i+\ell}^c, y_{j+r}^c) - q(x_{i+\ell}^c, y_{j+r}^c)) \chi(x_{i+\ell}^c, y_{j+r}^c)_{\tilde{\Omega}} \quad \text{subject to} \quad (5)$$

$$\nabla p(x_k^c, y_k^c) \cdot \mathbf{n}(x_k^c, y_k^c) = g(x_k^c, y_k^c), \quad (x_k^c, y_k^c) \in \Gamma \text{ collar points for } k = 1, \dots, N_{i+\frac{1}{2}, j+\frac{1}{2}}^c,$$

where g is the function that defines the Neumann boundary condition, i.e., $\nabla u \cdot \mathbf{n} = g$ on Γ .

- **First Order Accurate**

Wall BC Collar Points Constraints

- $\mathbf{v} \cdot \mathbf{n} = 0$
- $\partial_{\mathbf{n}} p = 0$
- $\partial_{\mathbf{n}}(\mathbf{v} \cdot \boldsymbol{\tau}) = 0$
- Larger system, similar blocks
- **First Order Accurate**

Exact Solution

$$\Omega = \{1 \leq \sqrt{x^2 + y^2} \leq 1.384\},$$

$$\rho(x, y) = 1,$$

$$u(x, y) = yf(\rho(x, y)),$$

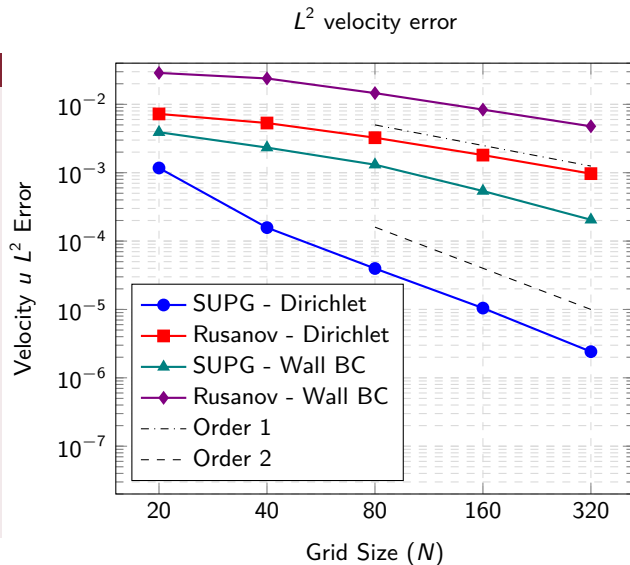
$$v(x, y) = -xf(\rho(x, y)),$$

$$r_0 = 2.5,$$

$$\rho(x, y) = \frac{\sqrt{x^2 + y^2}}{r_0},$$

$$f(\rho) = \gamma(1 + \cos(\pi\rho))^2,$$

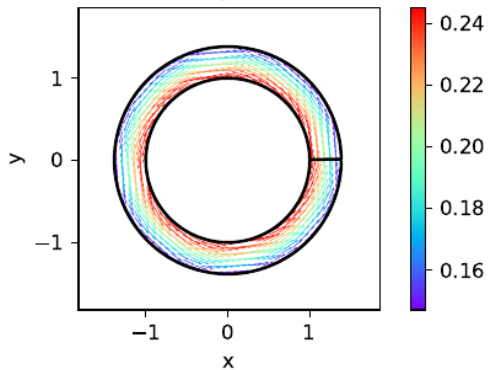
$$\gamma = \frac{12\pi\sqrt{0.981}}{r_0\sqrt{315\pi^2 - 2048}}.$$



Acoustic vortex in ring

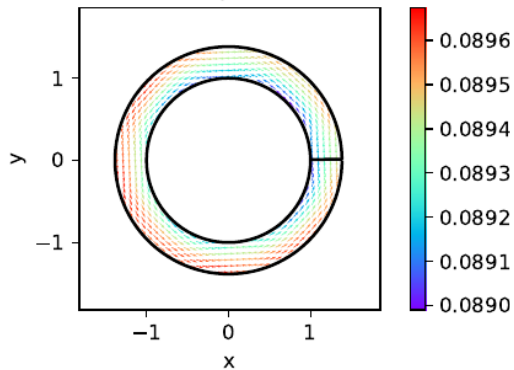
GF-SUPG-FV - GPM - Wall BC, $N_x = N_y = 40$

Velocity vector



GF-Rusanov-FV - GPM - Wall BC, $N_x = N_y = 40$

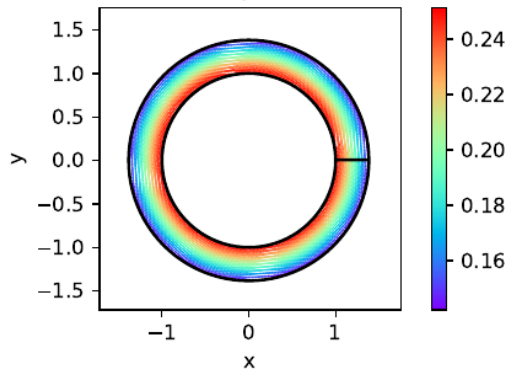
Velocity vector



Acoustic vortex in ring

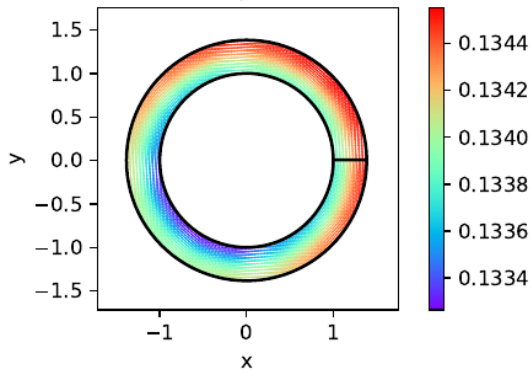
GF-SUPG-FV - GPM - Wall BC, $N_x = N_y = 80$

Velocity vector



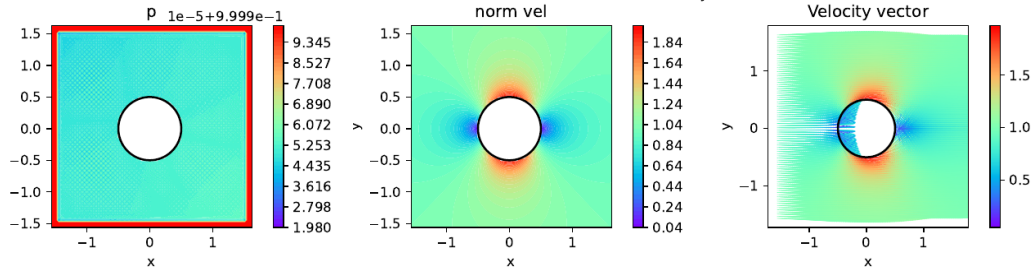
GF-Rusanov-FV - GPM - Wall BC, $N_x = N_y = 80$

Velocity vector

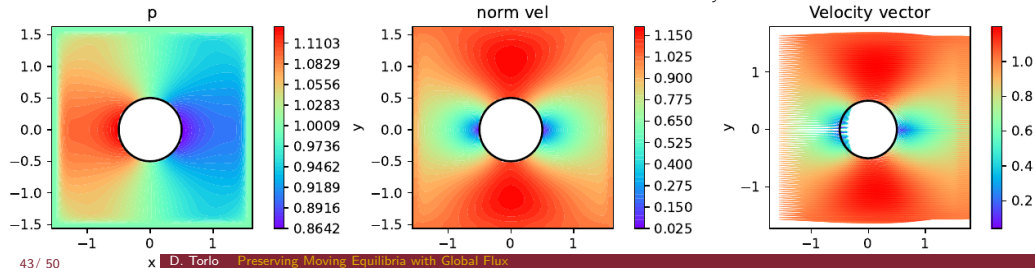


Potential Flow Around Cylinder

GF-SUPG-FV - GPM - Wall BC, $N_x = N_y = 80$

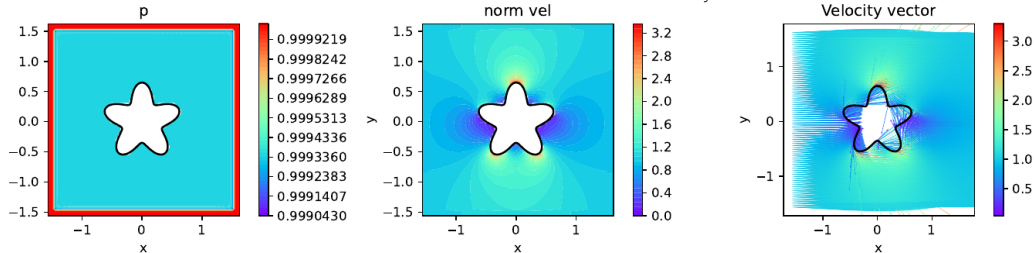


GF-Rusanov-FV - GPM - Wall BC, $N_x = N_y = 80$



Potential Flow Around Flower

GF-SUPG-FV - GPM - Wall BC, $N_x = N_y = 80$



GF-Rusanov-FV - GPM - Wall BC, $N_x = N_y = 80$

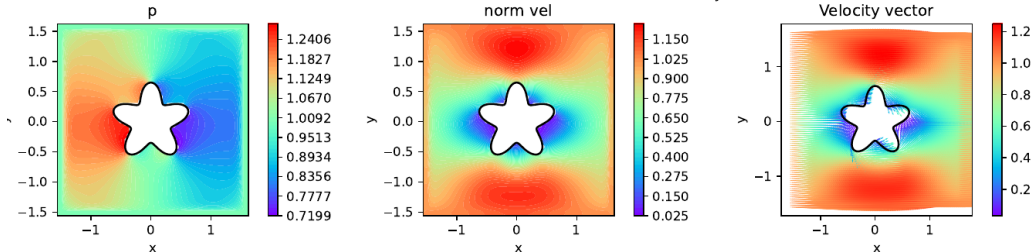


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Computational costs for reaching equilibria

Perturbation of equilibria

- To perturb the equilibrium, we need to know the **discrete** equilibrium
- Different strategies are possible: Optimization, Constrained Integration, etc.
- Simpler and most effective for complex cases: **Run the FOM until it reaches the discrete equilibrium and store the solution**
- Then add the perturbation on the top of that
- **Computational Costs are pretty high!**

Model order reduction

- Parametrized PDEs: we have a parameter that changes the equilibrium
- We want quickly a new equilibrium for a new parameter value
- The idea is to reduce the computational costs by reducing the number of degrees of freedom of the system (ROM)
- **Hope:** The ROM reaches the equilibria of the FOM with a reduced number of degrees of freedom

Classical POD+Galerkin

- FOM FEM SUPG GF:

$$\begin{cases} M_u \partial_t u + D_x p + \alpha D^x \partial_t p + \alpha D_x^x u + \alpha D_y^x v & = 0, \\ M_v \partial_t v + D_y p + \alpha D^y \partial_t p + \alpha D_x^y u + \alpha D_y^y v & = 0, \\ M_p \partial_t p + D_x u + D_y v + \alpha D^x \partial_t u + \alpha D_x^x p + \alpha D^y \partial_t v + \alpha D_y^y p & = 0, \end{cases} \quad (6)$$

- Compute SVD

- of u, v, p separately and truncate to get reduced basis V_u, V_v, V_p .
- of $(u, v), p$ and truncate to get reduced basis $V_{(u,v)}, V_p$.

- ROM with Galerkin projection:

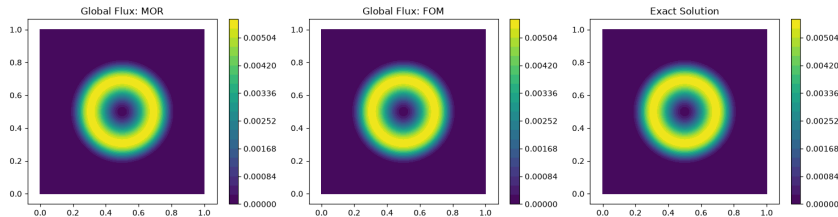
$$\begin{cases} V_u^T M_u V_u \partial_t \hat{u} + V_u^T D_x V_p \hat{p} + \alpha V_u^T D^x V_p \partial_t \hat{p} + \alpha V_u^T D_x^x V_u \hat{u} + \alpha V_u^T D_y^x V_v \hat{v} = 0, \\ V_v^T M_v V_v \partial_t \hat{v} + V_v^T D_y V_p \hat{p} + \alpha V_v^T D^y V_p \partial_t \hat{p} + \alpha V_v^T D_x^y V_u \hat{u} + \alpha V_v^T D_y^y V_v \hat{v} = 0, \\ V_p^T (M_p V_p \partial_t \hat{p} + D_x V_u \hat{u} + D_y V_v \hat{v} + \alpha D^x V_u \partial_t \hat{u} + \alpha D_x^x V_p \hat{p} + \alpha D^y V_v \partial_t \hat{v} + \alpha D_y^y V_p \hat{p}) = 0, \end{cases} \quad (7)$$

²with SUNHYPE group: Maria Kazolea, Ralph Lteif, Giuseppe Orlando, Alexandre Vieira, Michal Žeravý

Model Order Reduction preliminary results

Results

- The ROM is able to capture the dynamics of the FOM with a reduced number of degrees of freedom.
- Explicit ROM: speedup of 10-100x compared to FOM (depends on spatial discretization).
- Explicit ROM for divergence-free problems: $CFL \gg 1$ still stable, e.g. $CFL=5$, in general stable with same $CFL \Rightarrow$ speedup 150x.
- Implicit ROM vs Implicit FOM: speedup 20-1000x.
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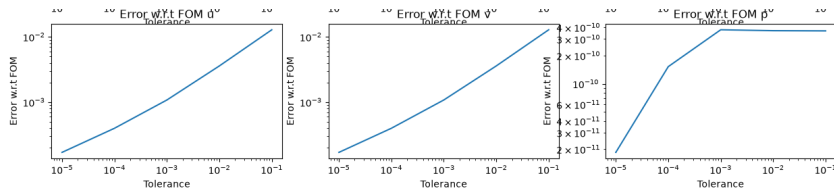


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Extensions and Perspectives

Summary

- Global Flux to preserve moving equilibria
- 1D integrate the source and unique flux
- 2D integrate f in y and g in x
- Some superconvergence in steady states
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- Extra accuracy in low Mach problems
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- Other methods: DG seems less trivial
- Shifted Boundary methods for FEM
- Other Riemann solver for corner problems
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THANKS!!